VOCCSYS

BY VULCAN INDUSTRIES INDIA

PRODUCT BROCHURE / VCS-SERIES

Clean condensers while the plant runs.

VOCCSYS is Vulcan Industries' patented Online Condenser Cleaning System — a continuous sponge-ball circulation loop that scrubs scale and biofilm from every condenser tube while the plant remains in operation. Zero shutdown. Zero chemicals. Measurable heat rate recovery.

0 hrs	1–3%	<12 mo
SHUTDOWN REQUIRED	HEAT RATE RECOVERY	TYPICAL PAYBACK

PATENTED · ONLINE CONDENSER CLEANING · ZERO SHUTDOWN

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PRODUCT BROCHURE · JULY 2026

voccsys.com

— 01 / THE SYSTEM

Fouling is silent.

So is our fix.

In every condenser, thin layers of scale, algae and biofilm form on the inner tube walls. You can’t see it — but you pay for it. Every micron of fouling degrades heat transfer, silently forcing higher back-pressure, higher fuel burn and quiet losses that add up to crores of rupees each year.

VOCCSYS eliminates it — continuously, mechanically, and without ever stopping your plant. Cleaning sponge balls circulate through every tube on every pass, scrubbing surfaces back to a mirror finish and keeping them there.

CONTINUOUS

Runs 24×7 with your condenser — no offline windows.

MECHANICAL

No chemicals, no additives, no environmental risk.

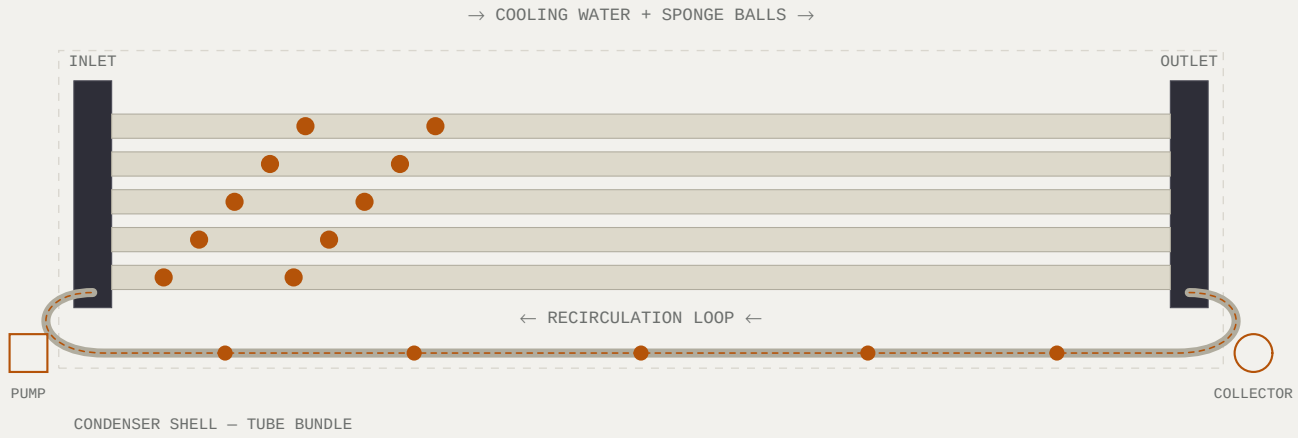
PROVEN

Deployed across power, HVAC, refining and steel plants.

— 02 / HOW IT WORKS

A closed loop.

Cleaning without pause.



01	**INJECT**	Sponge balls (slightly oversized vs tube ID) are injected into the cooling water inlet header via the VOCCSYS ball trap.
02	**SCRUB**	Balls flow through every condenser tube with the water, wiping the internal surface clean of scale and biofilm on every pass.
03	**COLLECT**	A downstream strainer captures 100% of the balls after the condenser without disturbing water flow.
04	**RECIRCULATE**	The recovery pump re-injects the collected balls back to the inlet, forming a closed, continuous cleaning loop.

— 03 / ROI & BENEFITS

Numbers that survive the finance meeting.

The value of VOCCSYS is neither theoretical nor marginal — it is measurable at the main condenser vacuum, at the turbine heat rate, and on your fuel invoice.

1–3%	HEAT RATE IMPROVEMENT	Continuous, measurable at the turbine.
6–12 mo	TYPICAL PAYBACK PERIOD	Across most large condensers.
40–60 %	REDUCTION IN FOULING LOSSES	Vs. offline mechanical cleaning cycles.
24/7	UNINTERRUPTED OPERATION	Zero shutdown, zero production loss.

Lower fuel & auxiliary power

A cleaner condenser drops back-pressure — every percentage point of vacuum recovered translates directly to fuel savings.

Stable heat rate

No more sawtooth degradation between shutdowns. Performance stays flat at design conditions.

No production loss

Cleaning happens with the plant running. Skip the offline cleaning windows entirely.

Zero chemicals

Fully mechanical scrubbing. No biocides, no acids, no environmental discharge concerns.

Extends tube life

Prevents pitting, under-deposit corrosion and biofouling — protecting your capital equipment.

Retrofit-ready

Designed to be installed on existing condensers with minimal piping modifications.

— 04 / TECHNICAL SPECIFICATIONS

Engineered to your condenser, not the other way around.

Every VOCCSYS unit is sized against your condenser's tube geometry, water chemistry and operating envelope. Below is a representative baseline.

CONFIGURATION

Model range	VCS-500 / VCS-1000 / VCS-2000 / Custom
Ball injection type	Continuous / batch programmable
Tube ID compatibility	18 mm – 32 mm
Tube materials	SS 304 / 316L, Titanium, Cu-Ni, Admiralty Brass

SPONGE BALLS

Base material	Natural rubber / EPDM foam
Ball grades	Standard, Abrasive, Extra-abrasive
Density	Marginally < cooling water
Typical life	1,500 – 3,000 hours per lot

SYSTEM

Control	PLC + HMI, remote SCADA-ready
Ball recovery	100% via mesh strainer / basket type
Piping	MS / SS with rubber lining, ANSI B16.5 flanges
Power supply	415 V, 3ϕ, 50 Hz (customisable)

COMPLIANCE

Design code	TEMA / HEI aligned
Instrumentation	ΔP, flow, turbidity, ball count sensors
Documentation	GA drawings, PID, IOM, O&M; manual
Warranty	12 months from commissioning

— 05 / INDUSTRIES SERVED

Wherever tubes foul, VOCCSYS pays back.

Thermal & Nuclear Power	Main and auxiliary condensers, closed-cycle cooling.	500 MW+
HVAC & District Cooling	Large chiller condensers in commercial and campus deployments.	5000+ TR
Petrochemical & Refining	Overhead condensers and heat exchangers in critical service.	24×7
Steel & Metals	Blast furnace, rolling mill and continuous casting cooling.	RETROFIT
Sugar & Paper	Vapour condensers and process heat exchangers.	SEASONAL
Chemical & Fertilizer	Process condensers with challenging water chemistry.	CORROSIVE

— 06 / CONTACT

Send us your condenser. We’ll size a system.

Share your plant details and one of our engineers will get back with a tailored technical proposal — typically within 2 working days.

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